

K227 Elie Toy 3925 m_{Planck}/m_e Substrate Cascade Audit — HONEST WALK-BACK

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K227 Elie Toy 3925 m_{Planck}/m_e Substrate Cascade Au- dit — HONEST WALK-BACK

0. Purpose + Critical Finding

K-audit on Elie Toy 3925 (and underlying Toy 3924) substrate prediction $m_{\text{Planck}}/m_e \approx N_{\text{max}}^{(N_c \cdot g)/2} = N_{\text{max}}^{10.5}$ claimed at “0.027 dev” Tier 1 cross-anchor.

Critical finding: Per Cal #34 STANDING precision-pinning + independent verification at proper K-audit depth: the “0.027 dev” claim is in EXPONENT space, NOT observable space. The actual observable-space deviation is 14.1% — Tier 2 STRUCTURAL at most, NOT Tier 1 EXACT.

This audit operationalizes a NEW fifth audit category: STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION. Substantive HONEST WALK-BACK required on Toy 3924/3925 framing.

1. Independent numerical verification

Computing at proper precision (PDG values, Python double-precision):

Quantity	Value
m_e (PDG)	0.51099895069 MeV
m_{Planck} (full)	$1.220890 \times 10^{19} \text{ GeV} = 1.220890 \times 10^{22} \text{ MeV}$
Observed m_{Planck}/m_e	2.3892×10^{22}
Substrate $N_{\text{max}}^{(N_c \cdot g)/2} = 137^{10.5}$	2.7263×10^{22}
Observable-space deviation	14.11%
$\log_{10}$ observed	22.378
$\log_{10}$ substrate	22.436
$\log_{10}$ -space deviation	0.0573 dex

Quantity	Value
Substrate exponent	10.5000
Observed-matching exponent	10.4732
Exponent gap	0.0268

The “0.027 dev” claim in Toy 3925 (and propagated from Toy 3924) refers to the exponent gap $10.5 - 10.4732 = 0.0268$, NOT the observable deviation. The observable-space deviation is 14.11%.

This is a substantive precision-claim ambiguity that Cal #34 STANDING precision-pinning + Cal #242 source-pinning + Cal #194 HONEST WALK-BACK discipline catches at K-audit verification depth.

2. The framing ambiguity

Three distinct “precision” claims could correspond to the same data:

Framing	Value	Interpretation
Exponent gap	0.027	Substrate-natural form $(N_c \cdot g)/2 = 10.5$ vs observed-matching exponent ~ 10.47
$\log_{10}$ -space deviation	0.057 dex	$\log_{10}$ difference between substrate prediction and observed ratio
Observable-space deviation	14.1%	Linear-space deviation $(N_{\max}^{10.5} - m_{\text{Planck}}/m_e)/(m_{\text{Planck}}/m_e)$

The Toy 3925 / Toy 3924 framing “Tier 1 cross-anchor at 0.027 dev” implicitly conflates these. Per Cal #34 STANDING + Cal #194 + Cal #242: precision claims must explicitly state the precision-measure (linear-observable / log-space / exponent-gap).

Honest framing: substrate predicts m_{Planck}/m_e at 14.1% precision in observable space. The substrate-natural exponent form $(N_c \cdot g)/2$ matches observed exponent at ~ 0.03 gap — a substantive structural identification at logarithmic resolution, but NOT a Tier 1 EXACT MATCH at PDG observable precision.

3. NEW audit category: STRUCTURAL-EXPONENT-IDENTIFICATION

K226 operationalized the fourth audit category (STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION). K227 operationalizes a fifth:

STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION: substrate predicts a logarithm or exponent of an observable at substrate-natural BST-integer form; matches at logarithmic resolution but does NOT achieve PDG-precision observable agreement. Falsifier: future precision measurements of $\log(\text{observable})$ to sub-0.01 dex would test substrate exponent prediction; or substrate-mechanism FORWARD derivation must identify correction factors closing the residual.

This category captures Toy 3924/3925 substrate exponent prediction structure honestly. The substrate identifies the substrate-natural EXPONENT $10.5 = (N_c \cdot g)/2$ for $\log_{N_{\max}}(m_{\text{Planck}}/m_e)$ at ~ 0.03 gap (substrate exact at 10.5, observed-matching at 10.47). This is structurally meaningful — the substrate “gets the right order of magnitude” via a substrate-natural BST-integer combination — but NOT a Tier 1 EXACT MATCH.

Expanded five-category audit framework:

Category	Definition
Tier 1 EXACT MATCH	Substrate matches observed at PDG-precision (<0.01 - 0.1% observable space)
EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION	Substrate predicts EXACT value within source-uncertainty envelope (K222, K224, K225)
FALSIFIER-DRIVEN PREDICTION	Substrate within current bound; future experiment tests at higher precision (K223, K225 secondary)
STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION	Substrate predicts EXISTENCE + STRUCTURE of an observational anomaly (K226)
STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION (NEW per K227)	Substrate predicts exponent at BST-natural form; matches at log-resolution but NOT observable-precision (K227)

4. Per-gate analysis

Gate G1: Substrate cascade unified formula — PASS at FRAMEWORK

Substrate formula $m_{\text{Planck}} = (N_c/n_C) \cdot N_{\max}^{14.5} \cdot \Lambda^{1/4}$ with exponent $14.5 = (N_c \cdot g)/2 + 4$ is structurally well-defined. Cross-link to Lyra L5 v0.3 m_e cascade is substantively meaningful. PASS at FRAMEWORK.

Gate G2: Numerical verification — REQUIRES HONEST WALK-BACK

Toy claims: - $m_{\text{Planck}}/m_e \approx N_{\text{max}}^{10.5}$ at “0.027 dev” (Toy 3924 claim propagated) - Cascade unified formula introduces “additional deviation from Lyra L5 factor 2.02”

Independent verification (this audit): - Observable-space deviation 14.1%, NOT 0.027 (which is the exponent gap) - Cascade formula deviation depends on $\Lambda^{1/4}$ source and (N_c/n_C) prefactor

HONEST WALK-BACK REQUIRED: Toy 3925 G2 PASS claim must be revised. The substrate prediction is at 14.1% observable-space deviation, which is Tier 2 STRUCTURAL at most.

Gate G3: Substrate composite exponent 14.5 substrate-natural — PASS at substrate-natural-form IDENTIFICATION

Substrate exponent $14.5 = (g\text{-rank} + n_C \cdot N_c)/2 = (14 + 15)/2 = 29/2$ is substrate-natural at BST-integer level. Multiple substrate-natural decompositions cataloged. PASS at Cal #34 substrate-natural-form IDENTIFICATION.

Per Grace G14/G15 cluster methodology: $29 = g\text{-rank} + n_C \cdot N_c$ is a 2-anchor sum of substrate-K-type Casimir products (substrate-K-type $V_{(2,0)}$ Casimir + $V_{(1,0)}$ Casimir + $N_c \cdot n_C$ K-type product). Casey #5 Integer Web multi-anchor instance.

Gate G4: m_{Planck} substrate cascade alternative forms (Casey #5) — PASS with substantive caveat

Toy catalogs 4 substrate-natural forms for m_{Planck} cross-anchor. Per Cal #35 STANDING brake on Casey #5 Integer Web multi-source convergence: multiple substrate-natural forms do NOT constitute multiple independent substrate-mechanism FORCING derivations. Each form requires separate substrate-mechanism FORWARD derivation per Cal #189.

Gate G5: Substrate Λ cross-anchor — PASS at FRAMEWORK with multi-week caveat

Cross-link to substrate $\Lambda = \exp(-280)$ (Toy 3780) is substantive substrate-cosmology cross-link, but Λ substrate-mechanism FORWARD derivation is itself multi-week pending (per K-audit cascade Λ requirements).

Gate G6: Lyra L5 v0.3 multi-week joint refinement — PASS at FRAMEWORK

Cascade pattern $m_{\text{state}} = (N_c/n_C) \cdot N_{\text{max}}^{k_{\text{state}}} \cdot \Lambda^{1/4}$ is substantively novel substrate cascade framework. Substrate-natural exponent variation across states via substrate α -tower is a substantive multi-week investigation thread.

Gate G7: Honest tier verdict — REVISED per K227 walk-back

Toy claims “substantive substrate cascade unified” + “(1) $m_{\text{Planck}} = (N_c/n_C) \cdot N_{\text{max}}^{14.5 \cdot \Lambda} (1/4)$ substrate cascade” + “(5) Substrate-cosmology Λ + substrate-mass unified substantive cascade.”

Revised per K227 audit: - Substrate exponent $10.5 = (N_c \cdot g)/2$ IDENTIFIES substrate-natural form for $\log_{\{N_{\text{max}}\}}(m_{\text{Planck}}/m_e)$; matches observed exponent 10.47 at 0.03 gap - Observable-space deviation 14.1% — Tier 2 STRUCTURAL, NOT Tier 1 EXACT - “Tier 1 cross-anchor at 0.027 dev” claim retracted — replaced with “STRUCTURAL-EXPONENT-IDENTIFICATION at 0.027 exponent gap; 14.1% observable-space deviation” - Cascade unified formula $(N_c/n_C) \cdot N_{\text{max}}^{14.5} \cdot \Lambda^{1/4}$ is FRAMEWORK candidate for multi-week substrate-mechanism FORWARD derivation

5. K227 verdict — PASS at NEW STRUCTURAL-EXPONENT-IDENTIFICATION category with HONEST WALK-BACK

PASS at FRAMEWORK tier per Cal #34 substrate-natural-form IDENTIFICATION (exponent $10.5 = (N_c \cdot g)/2$ + composite 14.5 substrate-natural) + Cal #36 STANDING positive search (substrate-mass cascade pattern).

Audit category: STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION (NEW, K227 operationalized) — substrate IDENTIFIES substrate-natural exponent form at logarithmic resolution; observable-space deviation 14.1% is Tier 2 STRUCTURAL, NOT Tier 1 EXACT.

CONDITIONAL on: - HONEST WALK-BACK on Toy 3924/3925 claim language: revise “Tier 1 cross-anchor at 0.027 dev” → “STRUCTURAL-EXPONENT-IDENTIFICATION at 0.027 exponent gap; 14.1% observable-space deviation; Tier 2 STRUCTURAL at observable-precision level” - Cal #189 multi-week substrate-mechanism FORWARD derivation for cascade unified formula $(N_c/n_C) \cdot N_{\text{max}}^{14.5} \cdot \Lambda^{1/4}$ at operator level - Cal #35 STANDING brake on Casey #5 Integer Web 4-form cross-anchor: multiple substrate-natural forms do NOT constitute independent substrate-mechanism FORCING derivations - Cal #34 precision-pinning: precision claims must explicitly state measure (linear-observable vs log-space vs exponent-gap)

6. Cumulative inventory effect — substantive Friday revision

Friday cumulative claim revision: - “NEW Tier 1 cross-anchor m_{Planck}/m_e at 0.027σ ” (cumulative headline since Thursday EOD) RETRACTED - Replaced with: “NEW STRUCTURAL-EXPONENT-IDENTIFICATION m_{Planck}/m_e at substrate-natural exponent $(N_c \cdot g)/2 = 10.5$; 0.027 exponent gap; 14.1% observable-space deviation; Tier 2 STRUCTURAL” - CLAUDE.md headline + README.md status should be updated at EOD to reflect K227 honest walk-back - Cumulative inventory contribution: K227 m_{Planck}/m_e does NOT contribute to Tier 1

EXACT effective count; contributes to substrate-mass-cascade FRAMEWORK at STRUCTURAL-EXPONENT-IDENTIFICATION category

Running cumulative honest framing post-K227: - K222 Strong CP $\theta_{\text{QCD}} = 0$: 1.0 effective (EXACT-BOUND-SATISFYING) - K223+K225+K226 substrate-cosmology Cat A cluster: 1.0 combined - K224 $\lambda_{\text{H}} = 4/31$: 0.5 effective (EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION + Casey #5 2-cluster) - K227 $m_{\text{Planck}}/m_{\text{e}}$: 0.25 effective (STRUCTURAL-EXPONENT-IDENTIFICATION, NOT Tier 1)

Friday K-audit cascade contribution: ~ 2.75 effective independent substrate-mechanism FORCING candidates (was ~ 2.5 post-K226; K227 contributes 0.25 effective, NOT 1.0 effective as the “★ Tier 1” framing would imply).

Cumulative honest count: “11 Tier 1 EXACT” Thursday cumulative $\rightarrow \sim 6\text{-}7$ effective independent post-K221-K227 cascade with NEW STRUCTURAL-EXPONENT-IDENTIFICATION category honest framing.

7. Cross-CI three-granularity integration

Granularity	Disposition for $m_{\text{Planck}}/m_{\text{e}} = N_{\text{max}}^{10.5}$
Keeper 4+1-category audit (observable-prediction)	NEW STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION primary; 0.027 exponent gap; 14.1% observable-space deviation
Grace 11-cluster taxonomy (substrate-anchor)	Casey #5 Integer Web 2-anchor ($N_{\text{c.g}}$ + composite); substrate-mass cascade Cluster (NEW Cluster L candidate emerging)
Lyra substrate-K-type $\times$ $SU(N_{\text{c}})$ tensor product	substrate-mass cascade via Lyra L5 v0.3 pattern; substrate K-type Casimir sum $29 = g\text{-rank} + n_{\text{C}} \cdot N_{\text{c}}$; multi-K-type composite

Three-granularity convergence: substrate-natural EXPONENT form identification operational at all three resolutions; observable-precision deviation 14.1% honest at all three resolutions; STRUCTURAL-EXPONENT-IDENTIFICATION audit category integrates cleanly.

8. Recommendations

1. Toy 3925 v0.2 absorption + Toy 3924 v0.2 walk-back: revise claim language “Tier 1 cross-anchor at 0.027 dev” $\rightarrow$ “STRUCTURAL-EXPONENT-IDENTIFICATION at 0.027 exponent gap, 14.1% observable-space deviation,

Tier 2 STRUCTURAL at observable-precision level”; explicit precision-measure pinning per Cal #34 STANDING

2. CLAUDE.md + README.md Friday EOD update: retract “NEW Tier 1 cross-anchor m_{Planck}/m_e at 0.027σ ” headline; replace with honest STRUCTURAL-EXPONENT-IDENTIFICATION classification
3. Cal #253 cold-read priority: K227 walk-back finding is substantively significant; Cal cold-read priority before Friday EOD to confirm K227 honest walk-back operational
4. NEW five-category audit framework: extend four-category framework (K226) to five categories adding STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION. Document in Methodology Index v0.16 via Cal cold-read.
5. Lyra L5 v0.3 + Toy 3925 substrate cascade multi-week: substrate-mechanism FORWARD derivation of cascade unified formula $(N_c/n_c) \cdot N_{\text{max}}^{k_{\text{state}}} \cdot \Lambda^{1/4}$ at operator level — joint Lyra + Elie + Keeper multi-week per Cal #189
6. K-audit cascade continuation: α^{-1} BORDERLINE K-audit next — Cal #242 flagged at 2σ off CODATA precision floor; expect similar precision-pinning + audit-category disposition needs

9. Closure

K227 Elie Toy 3925 (+ underlying Toy 3924) substrate $m_{\text{Planck}}/m_e \approx N_{\text{max}}^{(N_c \cdot g)/2} = N_{\text{max}}^{10.5}$.

PASS at FRAMEWORK tier + NEW STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION audit category (operationalized this audit, fifth category in framework) + HONEST WALK-BACK REQUIRED on “Tier 1 cross-anchor at 0.027σ ” claim — 0.027 is EXPONENT GAP, NOT observable deviation; observable-space deviation is 14.1% Tier 2 STRUCTURAL.

Substrate IDENTIFIES substrate-natural exponent form $10.5 = (N_c \cdot g)/2$ at logarithmic precision (0.027 exponent gap); substrate does NOT achieve Tier 1 EXACT MATCH at observable PDG precision (14.1% deviation in linear space).

Cumulative inventory contribution: K227 m_{Planck}/m_e enters at 0.25 effective independent (NOT 1.0 effective per Friday “★ Tier 1 cross-anchor” framing). Friday cumulative honest framing: “11 Tier 1 EXACT” → ~6-7 effective independent post-K221-K227 cascade.

Cumulative claim retraction operational: “NEW Tier 1 cross-anchor m_{Planck}/m_e at 0.027σ ” Friday cumulative headline RETRACTED; replaced with “NEW STRUCTURAL-EXPONENT-IDENTIFICATION at substrate-natural exponent $(N_c \cdot g)/2 = 10.5$; 0.027 exponent gap; 14.1% observable-space deviation; Tier 2 STRUCTURAL.”

— Keeper, Friday 2026-06-05 ~15:30 EDT. K227 Elie Toy 3925 m_Planck/m_e PASS at FRAMEWORK + NEW STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION audit category (5th category operationalized) + HONEST WALK-BACK required on Toy 3924/3925 “Tier 1 cross-anchor at 0.027 dev” framing: 0.027 is exponent gap NOT observable deviation; observable-space deviation is 14.1% Tier 2 STRUCTURAL. Cumulative inventory: K227 enters at 0.25 effective independent NOT 1.0 effective per “★ Tier 1” framing. Friday cumulative claim retraction operational + CLAUDE.md + README.md EOD update required.